

A Lean-Verified All-Density Exclusion for CMV Four-Arc Minimizers

Proof built, Lean-verified, and semantically accepted by
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Statement and geometric setting

Cañete, Miranda Jr., and Vittone study the isoperimetric problem for the lower-semicontinuous planar strip density and identify a symmetric four-arc family among the possible candidates. Their Conjecture 3.12 predicts that type-iv candidates are never minimizing; the relevant geometric classification and the type-iii/type-iv formulas occur in §3.2, especially Lemma 3.8 and equations (26)–(27) of the cited paper.

Fix $\lambda > 1$. On $\mathbb{R}^2$, set

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1, \end{cases} \quad A_\lambda(E) = \int_E f_\lambda d\mathcal{L}^2, \quad P_\lambda(E) = \int_{\partial E} f_\lambda d\mathcal{H}^1.$$

Here the boundary is the complete topological frontier of the Euclidean realization of the coordinate carrier, not a freely chosen list of arcs. The formal modeled regions carry the integrability hypotheses needed for the area and frontier integrals, and the admissible competitor type compares the candidate with empty, general, four-arc, and replacement carriers.

LEAN REFERENCES

```
CMVGeometry.lean StripDensity; WeightedArea; FrontierMeasure; WeightedPerimeter; FinitePerimeterRegion;
AdmissibleCompetitor; AdmissibleCompetitor.weightedArea_eq_carrier;
AdmissibleCompetitor.WeightedPerimeter
```

A modeled four-arc candidate has curvature and exterior half-angle satisfying

$$0 < h \leq 1, \quad 0 < \alpha < \frac{\pi}{2}, \quad \lambda \cos \alpha = h.$$

Its common radius is $R = 1/h$; the upper and lower exterior caps have chord length $2R \sin \alpha$ on the two density-one interfaces, and the strip core contributes two circular side arcs. The formal CMV predicate consists exactly of the density jump and Snell incidence. It does not assume an area formula, a competitor, or nonminimality. The incidence equation selects the principal inverse-cosine branch:

$$\alpha = \arccos(h/\lambda).$$

The analytic arguments use $0 < h < 1$, while the candidate constructor retains the finite closure value $h = 1$.

LEAN REFERENCES

```
FourArcCandidate.lean FourArcCandidate; FourArcCandidate.capChord; FourArcCandidate.stripCore;
FourArcCandidate.assembly; FourArcCandidate.SatisfiesCMVTypeIVHypotheses;
FourArcCandidate.alpha_eq_arccos
```

Theorem. For every $\lambda > 1$, every modeled regular type-iv four-arc candidate satisfying the formal CMV type-iv hypotheses fails to minimize weighted perimeter. Explicitly, the assertion

$$\forall F \text{ admissible, } A_\lambda(F) = A_\lambda(E) \Rightarrow P_\lambda(E) \leq P_\lambda(F)$$

is false for the modeled candidate E . The formal root has precisely this all-density modeled scope and includes the constructor's closure endpoint through the proved formula identities; it does not introduce a source-level normalization theorem.

LEAN REFERENCES

```
FourArcCandidate.lean FourArcCandidate.IsWeightedPerimeterMinimizer
UniversalStationaryPair.lean CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_density_gt_one
```

Scalar formulas and geometric realization

For the analytic comparison, write

$$B(x) = \lambda \arccos(x/\lambda) + \arcsin x, \quad D(x) = \frac{\sqrt{\lambda^2 - x^2}}{\lambda} - \sqrt{1 - x^2}, \quad q(h) = 2h - 1.$$

The scalar areas and perimeters are

$$A_4(h) = \frac{2(B(h) + hD(h))}{h^2}, \quad P_4(h) = \frac{4B(h)}{h},$$

$$A_3(h) = \frac{B(q(h)) + \pi/2 + (q(h) + 2)D(q(h))}{h^2}, \quad P_3(h) = \frac{2(B(q(h)) + \pi/2 + D(q(h)))}{h}.$$

The geometric type-iii carrier uses $0 < h < 1$; the type-iv candidate uses $0 < h \leq 1$. Formula values at a closure point are not silently treated as a regular carrier.

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeThreeShape; LeanSuffixAnalytic.typeThreeAngle;
LeanSuffixAnalytic.typeThreeDelta; LeanSuffixAnalytic.typeThreeArea;
LeanSuffixAnalytic.typeThreePerimeter; LeanSuffixAnalytic.typeFourAngle;
LeanSuffixAnalytic.typeFourDelta; LeanSuffixAnalytic.typeFourArea;
LeanSuffixAnalytic.typeFourPerimeter
```

For a regular type-iii competitor at curvature h , put

$$R = \frac{1}{h}, \quad \beta = \arccos q(h), \quad \alpha = \arccos(q(h)/\lambda), \quad w = R(\sin \alpha - \sin \beta).$$

The carrier is the union of a rectangle $[-w, w] \times [-1, 1]$, two strip-side circular segments, and an upper circular cap. Its complete frontier is the two strip-side arcs, the upper cap arc, and the bottom horizontal segment. The bottom segment is allowed to degenerate at $h = 1/2$; the upper cap is branch-complete, so major, semicircular, and minor cases are all represented.

LEAN REFERENCES

```
TypeThreeAssembly.lean TypeThreeAssembly; TypeThreeAssembly.coreCarrier; TypeThreeAssembly.outerCap;
TypeThreeAssembly.carrier; TypeThreeAssembly.bottomSegmentCarrier;
TypeThreeAssembly.bottomChord
```

A circular cap of radius R and half-central angle $\theta \in (0, \pi)$ has Euclidean area and arc length

$$R^2(\theta - \sin\theta\cos\theta) \quad \text{and} \quad 2R\theta,$$

respectively. The formal Hausdorff-measure identities identify these lengths with the Euclidean $\mathcal{H}^1$ -measure of the actual arc traces. Finite junction sets have zero one-dimensional measure, so the complete-frontier integrals are exactly the sums of the displayed component contributions.

LEAN REFERENCES

```
CMVGeometry.lean OneSidedCircularCap.volume_carrier
FrontierPerimeter.lean cap_arclength_eq_two_radius_theta; exact_euclidean_capTrace_hausdorffMeasure;
exact_euclidean_leftArcTrace_hausdorffMeasure;
exact_euclidean_rightArcTrace_hausdorffMeasure; fourArc_euclidean_frontier;
fourArc_frontier_weightedPerimeter_eq
TypeThreeAssembly.lean TypeThreeAssembly.exact_euclidean_bottomSegmentCarrier_hausdorffMeasure;
TypeThreeAssembly.euclidean_frontier;
TypeThreeAssembly.weightedPerimeter_eq_modeledWeightedPerimeter
```

The type-iii carrier is therefore a genuine measurable, integrable admissible competitor. Its geometric area and perimeter reduce to $A_3(h)$ and $P_3(h)$, while the candidate's complete-frontier area and perimeter reduce to $A_4(h)$ and $P_4(h)$.

LEAN REFERENCES

```
CMVSuffixModel.lean CMVSuffixModel.TypeThreeAssembly.scalarWeightedArea_eq_typeThreeArea;
CMVSuffixModel.TypeThreeAssembly.scalarWeightedPerimeter_eq_typeThreePerimeter;
CMVSuffixModel.FourArcCandidate.weightedArea_eq_typeFourArea;
CMVSuffixModel.FourArcCandidate.weightedPerimeter_eq_typeFourPerimeter
TypeThreeAssembly.lean TypeThreeAssembly.weightedArea_formula; TypeThreeAssembly.weightedPerimeter_formula;
TypeThreeAssembly.integrableOn_carrier; TypeThreeAssembly.integrable_frontierMeasure
```

Fold monotonicity and the stationary pair

On $0 < h < 1$, define

$$F_4(h) = 4 \left[h \left(\frac{1}{\sqrt{1-h^2}} - \frac{\lambda}{\sqrt{\lambda^2-h^2}} \right) - B(h) \right],$$

$$F_3(h) = 4h \left[\frac{1+q(h)}{\sqrt{1-q(h)^2}} - \frac{\lambda^2+q(h)}{\lambda\sqrt{\lambda^2-q(h)^2}} \right]$$

$$- 2 \left[B(q(h)) + \frac{\pi}{2} + D(q(h)) \right].$$

Differentiation gives

$$A_i(h) = \frac{F_i(h)}{h^3}, \quad P_i(h) = hA_i(h), \quad i \in \{3, 4\}.$$

Thus the fold zeros are stationary points of the area profiles; stationarity is used to choose a comparison pair, not assumed of the candidate being excluded.

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeThreeFold; LeanSuffixAnalytic.typeFourFold;
LeanSuffixAnalytic.hasDerivAt_typeThreeArea;
LeanSuffixAnalytic.hasDerivAt_typeFourArea;
LeanSuffixAnalytic.typeThree_variational_hasDerivAt;
LeanSuffixAnalytic.typeFour_variational_hasDerivAt
```

For $-1 < x < 1$, the radical-cube gap

$$K(x) = \frac{1}{(1-x^2)^{3/2}} - \frac{\lambda}{(\lambda^2-x^2)^{3/2}}$$

is positive whenever $\lambda > 1$. Consequently F_3 and F_4 are strictly increasing on their regular domains. In particular, A_3 has at most one fold transition, so its value on a compact interval is bounded by the larger endpoint value.

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.radicalCubeGap_pos; LeanSuffixAnalytic.hasDerivAt_typeThreeFold;
LeanSuffixAnalytic.hasDerivAt_typeFourFold;
LeanSuffixAnalytic.typeThreeFold_strictMonoOn;
LeanSuffixAnalytic.typeFourFold_strictMonoOn
SameCurvatureArea.lean LeanSuffixAnalytic.typeThreeArea_le_max_endpoints
```

To avoid evaluating the singular expression $F_4(1)$, clear its positive radical denominators with

$$C(h) = h(\sqrt{\lambda^2 - h^2} - \lambda\sqrt{1 - h^2}) - B(h)\sqrt{1 - h^2}\sqrt{\lambda^2 - h^2}.$$

This function is continuous through $h = 1$, agrees with a positive multiple of F_4 on $0 < h < 1$, is negative at $h = 9/10$, and is positive at $h = 1$. Hence some $s \in (9/10, 1)$ satisfies $F_4(s) = 0$. Strict fold monotonicity then gives

$$A_4(s) \leq A_4(h) \quad (0 < h \leq 1).$$

At the closure endpoint, the argument uses continuity of the finite area formula; it does not use an endpoint derivative of the singular raw fold.

LEAN REFERENCES

```
TypeFourStationaryRoot.lean LeanSuffixAnalytic.typeFourClearedFold;
LeanSuffixAnalytic.typeFourClearedFold_continuous;
LeanSuffixAnalytic.typeFourClearedFold_eq_fold;
LeanSuffixAnalytic.typeFourClearedFold_multiplier_pos;
LeanSuffixAnalytic.typeFourClearedFold_nine_tenths_neg;
LeanSuffixAnalytic.typeFourClearedFold_one_pos;
LeanSuffixAnalytic.typeFourFold_zero_exists
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeFourArea_fold_le
```

Equal-area comparison and the envelope

The same-curvature comparison is strict:

$$A_3(h) < A_4(h) \quad (\lambda > 1, \quad 0 < h \leq 1).$$

A support-function argument also gives, for $1/2 < h < 1$,

$$P_3(h) - hA_3(h) < P_4(h) - hA_4(h).$$

If $1/2 < r < h < 1$ and $A_3(r) = A_4(h) = V$, introduce the support-transfer function

$$H_V(x) = P_3(x) + x(V - A_3(x)).$$

The function H_V is nondecreasing from r to h . Combining this with the support inequality yields

$$P_3(r) < P_4(h),$$

and the increasing-fold argument yields $F_3(r) < 0$.

LEAN REFERENCES

```
SameCurvatureArea.lean LeanSuffixAnalytic.typeThreeArea_lt_typeFourArea; LeanSuffixAnalytic.typeFourSupport;
LeanSuffixAnalytic.typeFourSupport_hasDerivAt;
LeanSuffixAnalytic.typeFourSupport_concave0n;
LeanSuffixAnalytic.typeFourSupport_one_gt_halfPi;
LeanSuffixAnalytic.typeFourSupport_midpoint;
LeanSuffixAnalytic.sameHeight_support_identity;
LeanSuffixAnalytic.typeThree_support_lt_typeFour_support;
LeanSuffixAnalytic.typeThreeAreaSupport;
LeanSuffixAnalytic.typeThreeAreaSupport_hasDerivAt;
LeanSuffixAnalytic.typeThreePerimeter_lt_typeFourPerimeter_of_equalArea;
LeanSuffixAnalytic.typeThreeFold_neg_of_equalArea_of_lt
```

The endpoint values

$$A_3(1/2) = 2\pi(\lambda + 1), \quad A_4(1) < A_3(1/2)$$

bracket the stationary target $A_4(s)$. Continuity therefore produces $r \in (1/2, s)$ with

$$A_3(r) = A_4(s), \quad F_3(r) < 0, \quad P_3(r) < P_4(s).$$

Equivalently, there is a stationary equal-area pair with a strictly better type-iii member for every $\lambda > 1$.

LEAN REFERENCES

```
UniversalStationaryPair.lean LeanSuffixAnalytic.typeThreeArea_half;
LeanSuffixAnalytic.typeFourArea_one_lt_typeThreeArea_half;
LeanSuffixAnalytic.stationaryEqualAreaPair_of_typeFourFold_zero;
LeanSuffixAnalytic.stationaryEqualAreaPair_exists;
LeanSuffixAnalytic.stationaryEqualAreaPair_strictImprovement_exists
```

Fix such a pair (r, s) . Since $A_3(h) \rightarrow +\infty$ as $h \downarrow 0$, strict decrease on the negative-fold branch supplies, for every $h \in (0, 1]$, a unique descending root $\rho(h)$ satisfying

$$A_3(\rho(h)) = A_4(h), \quad F_3(\rho(h)) < 0.$$

The root obeys $\rho(h) < h$, is continuous on $(0, 1]$, and is differentiable in the interior with

$$\rho'(h) = \frac{A_4'(h)}{A_3'(\rho(h))}.$$

LEAN REFERENCES

```
LeanSuffixAnalytic.lean LeanSuffixAnalytic.typeThreeArea_tendsto_atRight_zero;
                        LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique;
                        LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique_of_stationary_pair
EqualAreaEnvelope.lean LeanSuffixAnalytic.typeThreeArea_strictAntiOn_Icc_of_fold_neg_right;
                        LeanSuffixAnalytic.equalAreaRoot_continuousOn_of_compact_unique;
                        LeanSuffixAnalytic.equalAreaRoot_hasDerivAt
SameCurvatureArea.lean LeanSuffixAnalytic.typeThreeArea_descending_root_lt
```

Define the equal-area perimeter gap

$$\Delta(h) = P_3(\rho(h)) - P_4(h).$$

The variational identities give

$$\Delta'(h) = (\rho(h) - h)A_4'(h).$$

The first factor is negative, while A_4' changes sign only at s . Thus Δ reaches its maximum at s , and

$$\Delta(h) \leq \Delta(s) = P_3(r) - P_4(s) < 0 \quad (0 < h \leq 1).$$

This is the all-curvature step: the stationary pair controls the envelope, but the excluded candidate may have any modeled curvature in its formal range.

LEAN REFERENCES

```
EqualAreaEnvelope.lean LeanSuffixAnalytic.equalAreaPerimeterGap;
                        LeanSuffixAnalytic.equalAreaPerimeterGap_hasDerivAt;
                        LeanSuffixAnalytic.equalAreaPerimeterGap_lt_fold;
                        LeanSuffixAnalytic.equalAreaPerimeterGap_le_fold;
                        LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope
```

For the candidate at curvature h , use the regular type-iii assembly at $\rho(h)$. The carrier realization gives exactly equal weighted area and strictly smaller complete-frontier weighted perimeter, so substituting it into the definition of modeled minimality is contradictory.

LEAN REFERENCES

```
CMVSuffixModel.lean CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree;
                    CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair
TypeThreeAssembly.lean FourArcCandidate.not_isWeightedPerimeterMinimizer_of_typeThree
```

Closing scope

This all-density theorem subsumes the earlier modeled cutoff $\lambda \geq 51/50$ and closes the entire modeled interval $1 < \lambda < 51/50$. The root concerns precisely modeled regular type-iv four-arc candidates satisfying the formal hypotheses, with the formal constructor's closure endpoint handled by the proved branch-complete identities. It does not claim the unqualified source-level CMV Conjecture 3.12.

Two separate bridges remain outside this root: normalization of every source-defined type-iv region into the formal candidate, and compatibility between CMV's reduced-boundary perimeter and the model's complete-frontier perimeter for the candidate and the constructed competitor. A universal classification of every arbitrary minimizer is

stronger than the bridge needed for this modeled theorem; the present result needs only the identified modeled candidate and one validated equal-area, lower-perimeter competitor.

LEAN REFERENCES

`UniversalStationaryPair.lean` `CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_density_gt_one`

Bibliography

A. Cañete, M. Miranda Jr., and D. Vittone, “Some Isoperimetric Problems in Planes with Density,” *The Journal of Geometric Analysis* **20** (2010), 243–290. Preprint arXiv:0906.1256; see §3.2, Lemma 3.8, Conjecture 3.12, and equations (26)–(27).